Giorgio Morales

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Education

Montana State University **Jan. 2022 – May 2025**
Ph.D., Computer Science. GPA: 3.97/4.0. *Bozeman, MT, USA*

Montana State University **Aug. 2019 – Dec. 2021**
M.Sc., Computer Science. GPA: 3.97/4.0. *Bozeman, MT, USA*

National University of Engineering **Aug. 2010 – Dec 2015**
B.S., Mechatronics Engineering. GPA: 3.5/4.0. *Lima, Peru*

Experience

Aston University **March 2026 – Present**
Aston AI Fellow *Birmingham, UK*

- Leading the *GUIDE* project, building human-in-the-loop software that distills complex black-box model behaviors into interpretable, data-driven mathematical equations using novel neuro-symbolic regression.
- Apply uncertainty quantification and interpretability to ensure transparency and reliability in automated discovery.

Université de Caen Normandie **Jan. 2025 – Feb. 2026**
Postdoctoral Researcher *Caen, NRM, France*

- Developed ML-based neutrino oscillation parameter estimation techniques using structured hierarchical transformers to accelerate large-scale physics simulations.
- Handled high-dimensional simulation datasets to extract robust statistical patterns and improve model prediction reliability.

Montana State University **Jan. 2021 – Dec. 2024**
Graduate Research Assistant *Bozeman, MT, USA*

- Designed and trained high-accuracy deep regression NNs and spatial models to predict crop yields and optimise agricultural resource allocation, providing uncertainty quantification, explainability components, and tailored clustering techniques.
- The developed techniques showed the potential to increase farmers' profits by up to $20,000 per field.
- Developed a web application in Flask to demonstrate the functionality of the prediction and optimization tools.
- Published 12 first-author journal and conference papers related to symbolic regression, prediction, and optimization in Precision Agriculture, uncertainty quantification, and explainable machine learning.

Montana State University **Aug. 2019 – Dec. 2020**
Graduate Teaching Assistant *Bozeman, MT, USA*

- Courses: Machine Learning (Fall 2020), Advanced Algorithm Topics (Fall 2020), Data Structures and Algorithms (Spring 2020), and Basic Data Structures and Algorithms (Fall 2019 and Summer 2020).

National University of Engineering – INICTEL-UNI **Nov. 2014 – July 2019**
Computer Vision Researcher *Lima, Peru*

- Led small teams and worked in multidisciplinary environments.
- Developed software to detect Mauritia Flexuosa palms in the Amazon using aerial images and drones. The project allowed the expansion of the monitored area by 200% and reduced the expedition costs by 500%.
- Developed software to segment clouds and shadows for the Space Agency of Peru (CONIDA) in high-resolution multispectral satellite images. The solution reduced the processing times by 1000%.

Selected Honors and Awards

Complete list: giorgiomorales.github.io/experience/#awards/

- Outstanding PhD Researcher (2025): MSU's Gianforte School of Computing Student Awards.
- Everyday Hero of Research (2025): MSU's Research Celebration – Celebrating Creativity.
- Seismic Monitoring and Analysis Challenge (2024): Awarded first prize in this competition hosted at ECML PKDD 2024.
- Cobleigh Endowment (2023, 2024): Scholarship for students of the Norm Asbjornson College of Engineering.
- Chunzi "Chris" Zhang (2023): Award for International Graduate Excellence in Research.
- William V. Benjamin (2021, 2022): Scholarship for students of the Norm Asbjornson College of Engineering.
- Southern Perú - ProUNI (2019): Graduate scholarship from the National University of Engineering.

Selected Publications

Complete list: giorgiomorales.github.io/publication/

Journal Articles

- **G. Morales** and A. Ekárt, "Exhaustive Canonical Skeletons Generation: A Formal Dataset for Decomposable Symbolic Regression," *IEEE Computational Intelligence Letters*, October 2026.
- **G. Morales** and J. Sheppard, "Decomposable Neural Symbolic Regression," *TMLR*, August 2026.
- **G. Morales** and J. Sheppard, "Dual Accuracy-Quality-Driven Neural Network for Prediction Interval Generation," *IEEE Transactions on Neural Networks and Learning Systems (TNNLS)*, December 2023.
- **G. Morales**, J. Sheppard, P. Hegedus, and B. Maxwell, "Improved Yield Prediction of Winter Wheat Using a Novel Two-Dimensional Deep Regression Neural Network Trained via Remote Sensing," *Sensors*, 23(1), 489, January 2023.
- **G. Morales**, J. Sheppard, R. Logan, and J. Shaw, "Hyperspectral Dimensionality Reduction Based on Inter-Band Redundancy Analysis and Greedy Spectral Selection," *Remote Sensing*, 13(18), 3649, September 2021.
- **G. Morales**, J. Sheppard, B. Scherrer, and J. Shaw, "Reduced-Cost Hyperspectral Convolutional Neural Networks," *Journal of Applied Remote Sensing*, 14(3), 036519 (2020), September 2020.
- **G. Morales**, G. Kemper, G. Sevillano, D. Arteaga, I. Ortega, J. Telles, "Automatic Segmentation of Mauritia flexuosa in Unmanned Aerial Vehicle (UAV) Imagery Using Deep Learning," *Forests*, 2018(9), 736, November 2018.

Refereed Conference Papers

- **G. Morales**, G. Lehaut, A. Vacheret, F. Jurie, and J. Fadili, "Neutrino Oscillation Parameter Estimation Using Structured Hierarchical Transformers," accepted to appear in *IEEE Int. Joint Conference on Neural Networks (IJCNN)*, June 2026.
- **G. Morales** and J. Sheppard, "Adaptive Sampling for Epistemic Uncertainty Reduction Using Prediction-Interval Generation Neural Networks," *AAAI Conference on Artificial Intelligence*, February 2025.
- **G. Morales** and J. Sheppard, "Univariate Skeleton Prediction in Multivariate Systems Using Transformers," *European Conf. on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML-PKDD)*, Sept. 2024.
- **G. Morales** and J. Sheppard, "Counterfactual Analysis of Neural Networks Used to Create Fertilizer Management Zones," *IEEE International Joint Conf. on Neural Networks (IJCNN)*, June 2024.
- **G. Morales** and J. Sheppard, "Counterfactual Explanations of Neural Network-Generated Response Curves," *IEEE International Joint Conf. on Neural Networks (IJCNN)*, June 2023.
- **G. Morales**, J. Sheppard, R. Logan, and J. Shaw, "Hyperspectral Band Selection for Multispectral Image Classification with Convolutional Networks," *IEEE Int. Joint Conf. on Neural Networks (IJCNN)*, July 2021.
- **G. Morales**, A. Ramírez, and J. Telles, "End-to-end Cloud Segmentation in High-Resolution Multispectral Satellite Imagery Using Deep Learning," *IEEE Int. Conf. on Electronics, Electrical Engineering and Computing (INTERCON)*, Aug. 2019.
- **G. Morales**, S. Huamán, and J. Telles, "Shadow Removal in High-Resolution Satellite Images Using Conditional Generative Adversarial Networks," *Int. Conf. on Information Management and Big Data (SIMBig)*, Feb. 2019.
- **G. Morales**, I. Salazar, J. Telles, and D. Díaz, "Detecting Violent Robberies in CCTV Videos Using Deep Learning," *Artificial Intelligence Applications and Innovations (AIAI)*, May 2019.

Refereed Workshop Papers

- **G. Morales** and J. Sheppard, "Learning Parametric Nitrogen Fertilizer Response Curves Using Neuro Symbolic Regression," *IEEE Congress on Evolutionary Computation (CEC): Workshop on Symbolic Regression and Equation Discovery*, June 2026.
- **G. Morales**, F. Jurie, and J. Fadili, "Towards Uncertainty Quantification in Generative Model Learning," *EurIPS Workshop: Epistemic Intelligence in Machine Learning*, December 2025.

Invited Papers

- **G. Morales** and J. Sheppard, "Two-dimensional Deep Regression for Early Yield Prediction of Winter Wheat," *SPIE Future Sensing Technologies*, November 2021.

Postgraduate Theses

- **G. Morales**, "Decomposable Neuro Symbolic Regression with Uncertainty Awareness," PhD dissertation, Gianforte School of Computing, Montana State University, 2025.
- **G. Morales**, "Towards Reduced-Cost Hyperspectral and Multispectral Image Classification," MS thesis, Gianforte School of Computing, Montana State University, 2021.

Language and Skills

- Spanish (native), English (full professional proficiency), Italian (B2), French (B1).
- Python, Java, C/C++, Pytorch, Tensorflow/Keras, Matlab, LaTeX.